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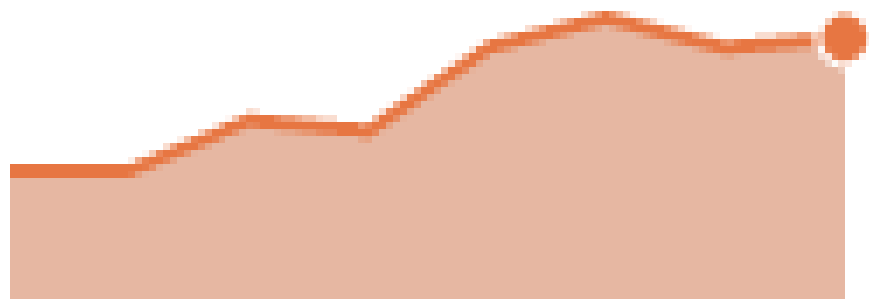
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Hybrid CNN–LSTM Integrated with Temporal Fusion Transformer for Accurate and Interpretable Stock Market Forecasting

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for accurate and interpretable stock

price forecasting. The CNN–LSTM captures short- and long-term dependencies, while the TFT

enhances temporal feature fusion and interpretability. Evaluated on Apple Inc. (AAPL) daily stock data over five years (2016–2020), the

proposed hybrid model achieved approximately 12% lower RMSE than a baseline LSTM model. Furthermore, a model-driven long/short

trading strategy based on the forecasts yielded a return of 80.7%, significantly outperforming the buy-and-hold benchmark return of 38%

over the same period. All results are reported before considering transaction costs. These findings demonstrate the proposed framework's

effectiveness in both predictive accuracy and real-world trading applicability.

Keywords:

time series prediction, TFT, CNN–LSTM hybrid model, stock market, financial time serie, trading strategy, explainable AI, multi step forecasting

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